

Rolling GHAR/GNNHAR-IV Scale Experiment: Local Verification Run

Volatility Research Pipeline

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1 Design

This verification run implements a Zhang-style rolling forecast design. For each forecast origin, the asset graph is recomputed from pre-origin returns using Graphical Lasso cross-validation. The forecasting block contains 22 trading days, and neural models use a pre-origin validation window. The target is 30-day close-to-close realized volatility; the IV extension adds 30-day mean implied volatility HAR lags. The model set is HAR, GHAR, GHAR2H, GHAR3H, HAR+IV, GHAR+IV, GHAR2H+IV, GHAR3H+IV, and GNNHAR1L/2L/3L with and without IV. GHAR2H and GHAR3H are linear multi-hop graph feature extensions; GNNHAR2L and GNNHAR3L are nonlinear multi-layer message passing models.

2 Run Scope

This is a local two-block verification run, not the final Colab long run. It is useful for checking data pairing, graph construction, model inclusion, and preliminary scale direction. A longer Colab run should increase the number of rolling blocks and neural epochs.

3 Universe Summary

Table 1: Rolling verification summary

Universe	Assets	Dates	Test days	Blocks	Best QLIKE	HAR QLIKE	GHAR+IV QLIKE
sp100	99	1073	44	2	GHAR+IV	0.00277137	0.00272525
sp500	449	1223	44	2	GHAR+IV	0.00576994	0.00544793

Table 2: Rolling graph construction

Universe	Assets	Blocks	Fallbacks	Median edges	Methods
sp100	99	2	0	881.5	glasso_cv:2
sp500	449	2	0	2558.5	glasso_cv:2

Table 3: Selected QLIKE gains

Universe	Comparison	QLIKE gain
sp100	Graph gain without IV: GHAR vs HAR	-0.22%
sp100	Graph gain with IV: GHAR+IV vs HAR+IV	0.03%
sp100	IV gain in HAR: HAR+IV vs HAR	1.63%
sp100	IV gain in GHAR: GHAR+IV vs GHAR	1.88%
sp500	Graph gain without IV: GHAR vs HAR	0.35%
sp500	Graph gain with IV: GHAR+IV vs HAR+IV	0.09%
sp500	IV gain in HAR: HAR+IV vs HAR	5.50%
sp500	IV gain in GHAR: GHAR+IV vs GHAR	5.25%

4 Graph Audit

5 Scale Gains

6 Model Rankings

6.1 SP100

6.2 SP500

7 Interpretation

In this verification run, the graph-only GHAR gain is small. For S&P100 it is slightly negative by QLIKE, while for S&P500 it is positive but only about 0.35%. The IV channel is much stronger and scales up: IV gain in GHAR is about 1.88% for S&P100 and 5.25% for S&P500. Linear multi-hop GHAR does not dominate one-hop GHAR in these two blocks; GHAR+IV remains the best QLIKE model in both universes. The neural GNNHAR models underperform in this short local run, so their ranking should not be treated as final without the longer Colab run and broader tuning.

Table 4: sp100 top models by QLIKE

Model	Adjacency	QLIKE	MSE
GHAR+IV	Rolling GLASSO-1hop	0.00272525	6.10924
HAR+IV	Identity	0.00272614	6.10773
GHAR2H+IV	Rolling GLASSO-2hop	0.00272798	6.11961
GHAR3H+IV	Rolling GLASSO-3hop	0.00273196	6.11973
HAR	Identity	0.00277137	6.31694
GHAR	Rolling GLASSO-1hop	0.00277747	6.32539
GHAR2H	Rolling GLASSO-2hop	0.00278154	6.33722
GHAR3H	Rolling GLASSO-3hop	0.00278165	6.31748

Table 5: sp500 top models by QLIKE

Model	Adjacency	QLIKE	MSE
GHAR+IV	Rolling GLASSO-1hop	0.00544793	16.472
GHAR2H+IV	Rolling GLASSO-2hop	0.00544939	16.4914
GHAR3H+IV	Rolling GLASSO-3hop	0.00544994	16.4907
HAR+IV	Identity	0.00545277	16.4541
GHAR2H	Rolling GLASSO-2hop	0.00574786	17.2072
GHAR	Rolling GLASSO-1hop	0.00574991	17.1616
GHAR3H	Rolling GLASSO-3hop	0.00575064	17.2076
HAR	Identity	0.00576994	17.1461